McKenzie McCain

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EDUCATION

Texas A&M University – College Station, TX *May 2027*
B.S. in Bionics and Robotics, Interdisciplinary Track
- **Minors:** Biomedical Engineering (biomechanics) **Cumulative GPA:** 3.96/4.00
- **Organizations:** Biomedical Engineering Society (BMES), Engineers without Borders (EWB), Aggie Blossoms

Relevant Courses: Geometric Modeling, Engineering & Newtonian Mechanics, Calculus I-III, Differential Equations
Licenses & Certifications: Additive Manufacturing: Optimizing 3D Prints, Learning 3D Printing

PROJECTS

EMG-Controlled Prosthetic Hand | *T.U.R.T.L.E. Robotics OLSN Project* | College Station, TX *May 2025 – Present*
- Collaborated with the research and development of an electromyographic-controlled prosthetic hand for a local boy
- Utilized SolidWorks to design and print a scan for the client's arm, manufacture a socket, and create a mechanical hand system

Wireless Robot Controller *May 2025 – Present*
- Designed and programmed a custom wireless controller with ESP32, OLED display, joysticks, and buttons to control the robot from the T.U.R.T.L.E. Hatchling Competition.
- Applied embedded systems programming with ESP-NOW wireless protocol to integrate motor, pulley, and servo actuation with low-latency communication and responsive user interface.

Elderly Fidget Device | *Geometric Modeling Competition* | College Station, TX *August 2024 – November 2024*
- Designed and modeled an ergonomic fidget device for the elderly using SolidWorks, developing greater proficiency in CAD design and rapid prototyping/additive manufacturing.
- Competed in a geometric modeling competition, winning '*best prototype*' among 15+ competing teams

ACTIVITIES

T.U.R.T.L.E. Robotics | College Station, TX
Hatchling Director *April 2025 – Present*
- Oversaw 100+ college students from various organizations and taught topics such as soldering, embedded programming, Solidworks/computer-aided design, and mentored individual competition teams

Hatchling Member *December 2024 – April 2025*
- Designed a robot from scratch, utilizing Solidworks to 3-D model and print robotic components–placed 2nd against 21 competitors
- Worked with electronics (ESP32, L298n, DC-motors, and SG90 Micro Servos) to configure a custom electronic/mechanical system; utilized Arduino to program PWM motor control and mechanism functions to stack 2.5in blocks

Aggie Blossoms | College Station, TX *September 2024 – Present*
- Collaborated in planning events that promote empowerment and support among women students
- Served my community through social activities and community events

Biomedical Engineering Society | College Station, TX *January 2025 – Present*
- Networked with companies in the BMEN industry during weekly meetings; Improved professional skills and participated in mentorship program events

Freshman Leaders in New Technology (FLiNT) | College Station, TX *October 2023 – April 2024*
- Organized campus events promoting student engagement and community involvement
- Collaborated with peers on projects, fostering leadership growth and personal development

EXPERIENCE

Rex Academy | *Programming Summer Camp Instructor (Contract)* | Southlake, TX *June 2025 – July 2025*
- Developed lesson plans in the Rex Academy TLP; taught Scratch Jr. (K-2) and 2D Game Programming (3-5) with Flowlab
- Managed student accounts/devices and classroom setup; created a quick-start login guide and resolved bug/connectivity issues.
- Supervised and coached students to produce multiple playable mini-games and scene animations by week's end

Cinepolis Cinema | *Cast Member* | Euless, TX *March 2022 – July 2022*
- Provided high-quality customer service to over 200 patrons daily, performed opening/closing duties, and managed inventory supplies

Colleyville Baseball Association | *Cashier and Concessions Supplier* | Colleyville, TX *August 2019 – May 2021*
- Provided excellent customer service, assisting 250+ patrons and accurately balancing the register, processing up to $1000 per shift
- Maintained a welcoming environment and managed inventory/restocking of supplies, handling up to $600 deliveries

SKILLS

Tools & Platforms: SolidWorks, Autodesk (Fusion 360, Inventor), Visual Studio Code, Git/Github, Microsoft Office (Word, Excel, PowerPoint), Google Workspace (Docs, Sheets, Slides)
Technical: CAD/Design (Finite Element Analysis, 3-D Printing, Additive Manufacturing), **Embedded Systems** (Circuit Design, Microcontroller Programming (Embedded C)), **Robotics & Controls** (Controls Systems), **Programming** (Python, C/C++, Java)
Engineering & Analytical: Product Design/Development, Technical Report Writing, Mechanical System Design, Problem Solving